13

Veno-Arterial ECMO Physiology

Let's now move our discussion to the physiology of veno-arterial extracorporeal membrane oxygenation (VA ECMO). As you will come to appreciate, VA ECMO is a very different type of support than veno-venous extracorporeal membrane oxygenation (VV ECMO). These next few chapters will equip you with a much better sense of some of the subtleties related to this mode.

Similar to VV ECMO, VA ECMO involves a drainage cannula in the venous system. The blood is run through the same blood pump, membrane oxygenator, and circuit. However, the return cannula is implanted into the arterial system. As opposed to VV ECMO, which was dependent on the native cardiac output, the blood flow for VA ECMO translates to some of the overall cardiac output (Fig. 13.1).

There are limitations and rationale to the use of VA ECMO support. Let's start to contextualize these by introducing the vicious cycle of cardiogenic shock.

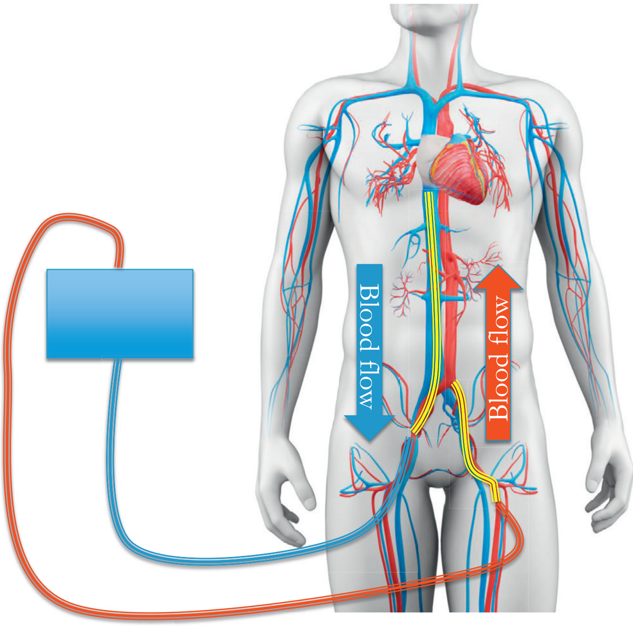

FIG. 13.1 Veno-arterial ECMO. (Modified from SciePro/Shutterstock.com.)

VICIOUS CYCLE OF CARDIOGENIC SHOCK

Cardiogenic shock can be one of the most devastating forms of shock, with mortality that can be as high as 85%. Worse yet, it is rapid, with a large proportion of patients dying within hours. You can see why we spent so much time talking about recognition, diagnosis, and identification of shock in the Physiology section. You will see that the reason for the rapid and devastating manifestation is the vicious cycle that is triggered.

The usual way of thinking is that decreased cardiac output leads to decreased perfusion, which leads to decompensation (Fig. 13.2)

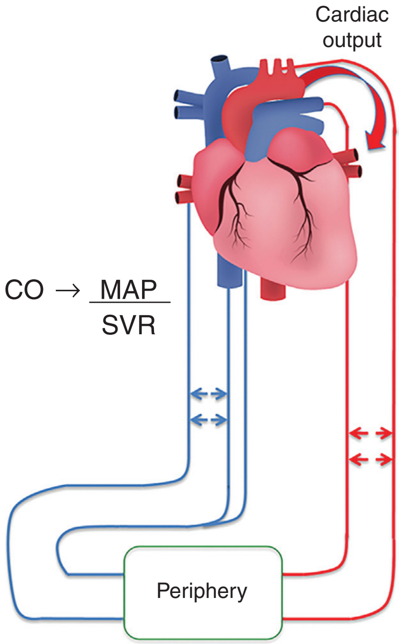

FIG. 13.2 Decreased cardiac output leading to decreased overall perfusion. *MAP*, Mean arterial pressure; *SVR*, systemic vascular resistance. (Modified from Ody_Stocker/Shutterstock.com.)

While this is true, there is much more at play here, with multiple mechanisms of decompensation that re-inforce and compound each other. Let's explore some of these effects.

Decreased Cardiac Output

Cardiac output is the driver of perfusion pressure, which is ultimately required for the forward flow of blood and delivery of oxygen. An insult to the cardiac function, whether ischemic due to myocardial ischemia, inflammatory as in myocarditis, obstructive as in pulmonary embolism or tamponade, or structural as in valvular/septal rupture, ultimately leads to a drop in cardiac output. As this output drops, the pressure head required to drive the forward flow of blood becomes progressively difficult to maintain, leading to a compensation in systemic vascular resistance (SVR) to maintain the forward flow of blood.

Decreased Coronary Perfusion

A drop in blood pressure affects the perfusion of all organs, but a decrease to the coronary blood vessels can be especially devastating because it serves to further worsen myocardial ischemia and further drop cardiac output.

Increased Left Ventricular End Diastolic Pressures (LVEDP)

Elevated SVR maintains the pressure needed to drive the forward flow of blood, much like putting your thumb over a hose may allow water to reach further if the flow of water decreases. However, this comes at a price – worsening afterload. As afterload increases, the left ventricular output starts to fall, which leads to less blood ejected with each heartbeat, and increasing the blood that remains in the ventricular cavity at the end of diastole (Fig. 13.3).

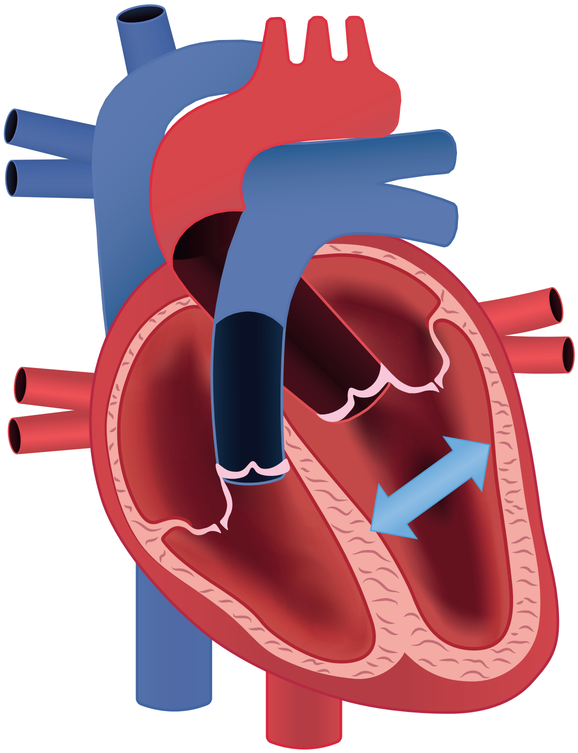

FIG. 13.3 Effect of afterload on failing left ventricle leading to increased left ventricular end diastolic pressure. (Modified from Ody_Stocker/Shutterstock.com.)

This elevated pressure has several adverse effects, especially in a failing and potentially ischemic left ventricle:

- Higher pressures can compress cardiac muscle causing wall stress leading to worsening ischemia
- Higher left ventricular pressures can exert higher pressures on blood returning from the lungs which can lead to pulmonary edema
- Pulmonary edema can lead to further hypoxia worsening overall oxygen delivery to the heart tissue which can further drop the cardiac output

You can see how this contributes to a downward spiral, with one adverse effect compounding on another.

Acidosis

As oxygen delivery worsens in the setting of hypoxia and decreased cardiac output, the cells throughout the body have less available oxygen to carry out aerobic metabolism. This causes them to shift to anaerobic metabolism, with worsening acidosis ensuing. While this process temporarily allows for the generation of energy at the cellular level, acidosis can precipitate vasodilation, worsening both venous return and shock.

Acute Kidney Injury

Amongst the first organs affected by worsening shock and oxygen delivery are the kidneys. The response can be adaptive initially, causing fluid retention and improvement in preload, but injury can ensue, leading to further fluid retention and decreased clearance. This can worsen acidosis and lead to volume overload with a host of adverse effect to include worsening pulmonary edema.

Right Ventricular Distension

The other adverse effect of volume overload is right ventricular distension. Remember that preload only helps the right ventricle to a point, after which there is little more stretch that can be accommodated by the less muscular ventricle. Further increase in right ventricular filling pressures can cause a precipitous decline in the output of the failing right ventricle and can decrease left ventricular output as the distended right ventricle begins to compress on the left ventricle (Fig. 13.4).

This can worsen if excessive fluids are administered to counteract the shock in this case.

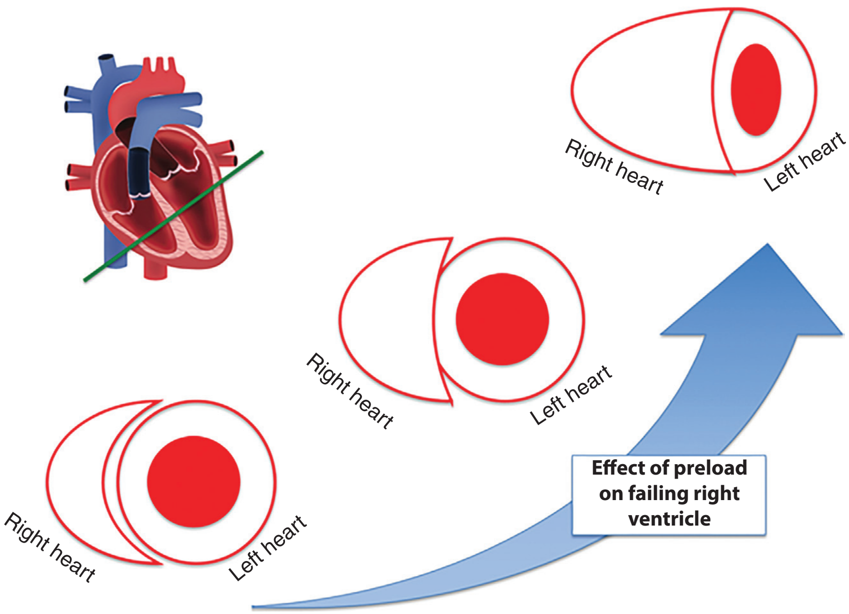

FIG. 13.4 The adverse effect of escalating preload on a failing right ventricle. (Modified from Ody_Stocker/Shutterstock.com.)

Increased Right Ventricular Afterload

The right ventricle is very afterload sensitive and will worsen in the following scenarios: hypoxia, hypercapnia, acidosis, hypothermia, decreased pulmonary compliance, and excessive pressor administration. As shock and pulmonary edema progress, all of these combine to worsen the pulmonary pressures and resistance that the right ventricle must work against, further decreasing output.

Summary of Vicious Cycle of Cardiogenic Shock

When taken together, we can see how these components of the vicious cycle of cardiogenic shock can compound on each other leading to a rapid decompensation. Although these components can all seem destructive, by separating out into their components we can have a better approach to all that is going on with a patient in cardiogenic shock, give a better framework to assess the overall effect of interventions, and define the role of mechanical circulatory support such as ECMO (Fig. 13.5)

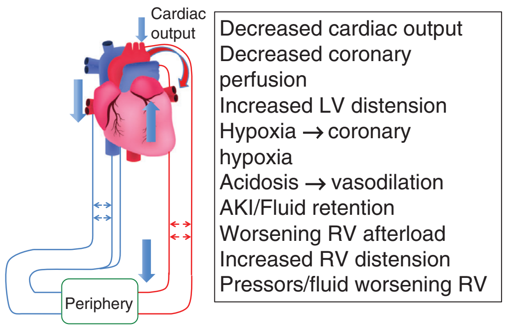

FIG. 13.5 Vicious cycle of cardiogenic shock. *AKI*, Acute kidney injury; *LV*, left ventricular; *RV*, right ventricular. (Modified from Ody_Stocker/Shutterstock.com.)

RATIONALE OF VA ECMO IN CARDIOGENIC SHOCK: EFFECT ON THE VICIOUS CYCLE

Let's now introduce VA ECMO and define how it is going to interact with this vicious cycle. There are two flavors of VA ECMO, central VA ECMO, where the cannulas are inserted directly onto the great blood vessels/heart, and peripheral VA ECMO, where the cannulas are inserted into the femoral blood vessels. We will discuss the physiology of central VA ECMO in Chapter 15, so for this chapter, we will focus on peripheral VA ECMO.

The blood flow is unique in VA ECMO, differing from VV ECMO in that blood is drained out of the venous system via a drainage cannula implanted into the inferior vena cava (IVC), and returned into the arterial system in a retrograde fashion against the normal flow of blood (Fig. 13.6).

The dynamics of this blood flow pattern are unique. Let's explore the effects this may have in the context of cardiogenic shock.

Effect #1: Support MAP

The effect of blood flow in ECMO is profound. In contrast to VV ECMO, where ECMO oxygen delivery is dependent on the native cardiac output, it is additive in VA ECMO, such that the total cardiac output is the summation of the native cardiac output and the ECMO blood flow (Fig. 13.7)

Since cardiac output is the mean arterial pressure (MAP)/SVR, an increase in overall output/perfusion translates into a higher MAP, or, importantly, allows for a decrease in SVR.

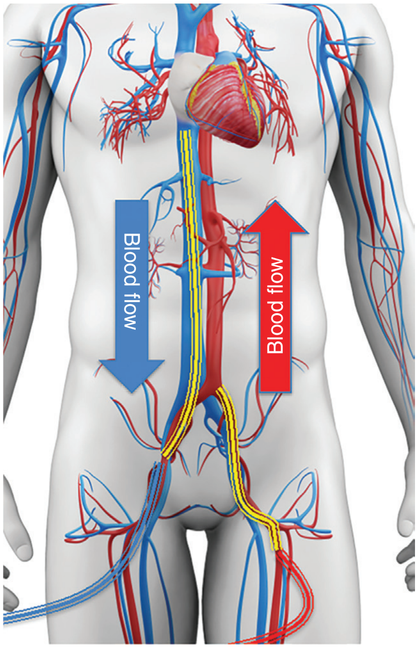

FIG. 13.6 Schematic of blood flow in peripheral VA ECMO. (Modified from SciePro/Shutterstock.com.)

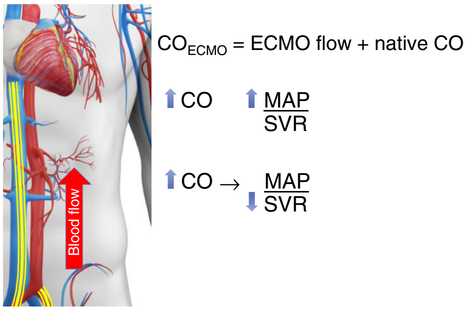

FIG. 13.7 Impact of VA ECMO flow on overall perfusion pressure and systemic vascular resistance. (Modified from SciePro/Shutterstock.com.)

Effect #2: Decrease SVR

Maintaining an adequate MAP at a lower SVR is essential. The whole purpose of MAP is to maintain the pressure needed to perfuse the end organs and importantly maintain the vascular pressure head.

What is meant by vascular pressure head? Since there is more compliance in the venous system than in the arterial system, there exists a forward drive of blood from higher pressure to lower pressure, as illustrated in Fig. 13.8.

This pressure head is essential for the forward flow of blood and for sustaining the return of blood flow back to the heart from the venous system. Therefore, as cardiac output drops,

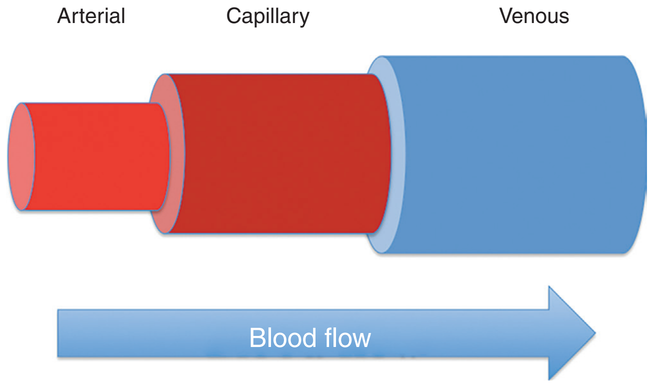

FIG. 13.8 The vascular pressure head as a mechanism for driving the forward flow of blood.

maintaining MAP with pressors through an increase in SVR only translates into an improvement in cardiac output from improvement in vascular pressure head leading to better venous return and ultimately better forward flow. If not, then increasing MAP with escalating pressors may only function to worsen overall perfusion.

Improvement in cardiac output with ECMO allows for a decrease in SVR needed to maintain this forward flow, improving blood flow to the end organs. Additionally, reducing the dose of pressors mitigates the adverse effects that high-dose pressors can have on cardiac output, such as worsening pulmonary vascular resistance/right ventricular afterload.

Effect #3: Perfuse Visceral Organs

Imagine the normal flow of blood out of the heart. It flows past the aortic valve into the ascending aorta and down the descending aorta. Within the next 12–18 inches it perfuses the liver, kidneys, intestines and mesentery, and entire abdomen. The lack of blood flow in cardiogenic shock causes a decline in blood flow to these essential organs as indicated by worsening lactate levels, liver enzymes, and renal function tests.

Now let's imagine what happens in ECMO. You have a column of oxygenated blood that is directed through this entire path. The result can be a rapid improvement in the perfusion of all of these organs (Fig. 13.9).

As perfusion is restored, you may witness a rapid improvement in urine output (minutes), lactate clearance/gut function (hours), and liver enzymes (days). The degree of this improvement is dependent on the extent that the injury was due to lack of perfusion rather than permanent damage. Patients with the latter fall into the "too far gone category" where ECMO may be of less benefit. However, teasing out where that line exists can be challenging when selecting patients for ECMO.

The result of these changes is a potential reversal of the vasodilatory effect of acidosis/renal failure/liver injury.

Effect #4: Provide Oxygen

Just as hypoxia exacerbates the vicious cycle, worsening delivery of oxygen to the coronary circulation, ECMO provides oxygen to the hypoxic patient in cardiogenic shock, mitigating this adverse effect. Oxygenated blood from the ECMO circuit can improve the overall delivery of oxygen which can translate to improved perfusion to multiple organ systems. However, this may be potentiated by the nature of the retrograde circulation.

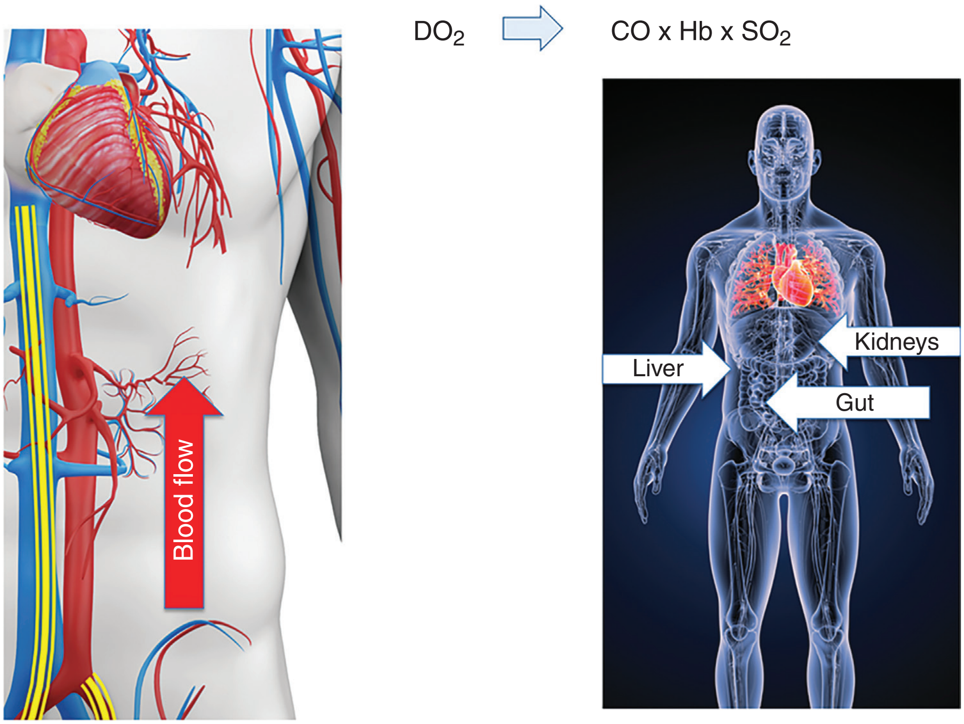

FIG. 13.9 The impact of ECMO blood flow on visceral organ perfusion. (From SciePro/Shutterstock.com and Sciencepics/Shutterstock.com.)

Effect #5: Support the Right Ventricle

By now you have likely developed a newfound respect for the role of the right ventricle in shock. While the left ventricle is responsible for the flow of blood to the rest of the body, it is completely dependent on the right ventricle, as the right ventricular cardiac output is the primary contributor to left ventricular preload. Thus the effects of right ventricular preload and afterload can impact the perfusion of the entire body.

Right ventricular afterload can be crushing. Causes of elevated pulmonary pressures include high left-sided pressures, hypoxia, poor pulmonary compliance, and hypercapnia. Improvement in oxygenation/CO_2 clearance from the ECMO circuit as well as improved acidosis from splanchnic perfusion can help to improve right ventricular afterload.

Additionally, in contrast to the effect of VV ECMO, the effect of VA ECMO on the right ventricle extends beyond afterload reduction. This is because any blood flow to the ECMO circuit represents blood that is drained from the right side, decompressing the right heart. If the right heart is over-distended to the point that it is impeding the filling of the left side of the heart, then this flow of blood will function to further improve overall perfusion (Fig. 13.10).

Effect # 6: Retrograde Blood Flow and the Left Ventricle

The flow of blood from the ECMO circuit exerts the opposite effect on the left ventricle in peripheral ECMO. While blood is drained away from the right ventricle in the venous circulation, it flows against the left ventricle since it is directed from the femoral artery up the aorta into the arterial circulation. Consequently, this flow and direction of blood *increases* left ventricular afterload and works against left ventricular cardiac output (Fig. 13.11).

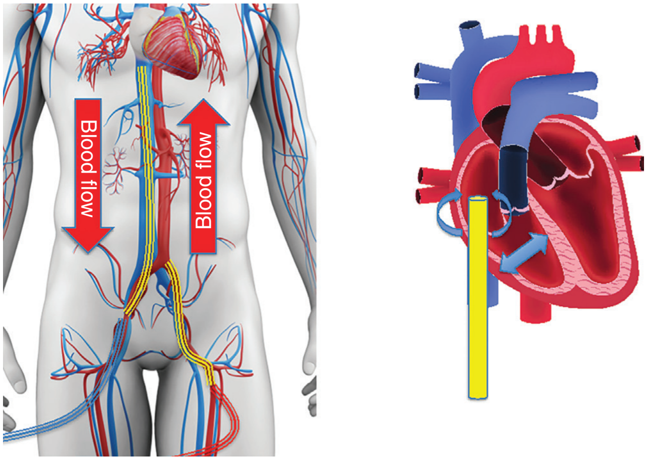

FIG. 13.10 VA ECMO as a mechanism of supporting the right ventricle. (From SciePro/Shutterstock.com and Ody_Stocker/Shutterstock.com.)

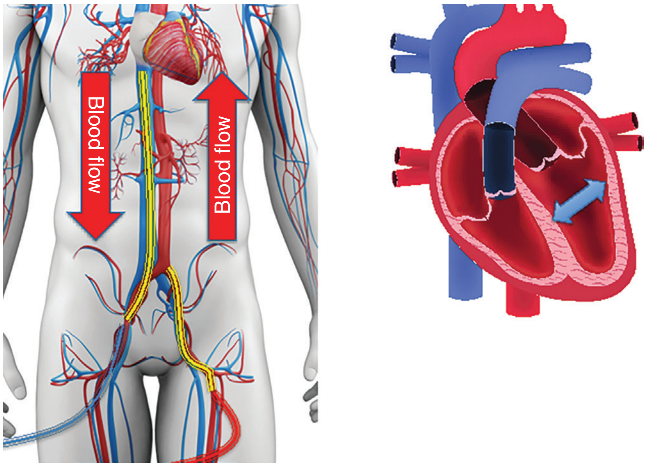

FIG. 13.11 The impact of retrograde flow on a failing left ventricle. (From SciePro/Shutterstock.com and Ody_Stocker/Shutterstock.com.)

Having an appreciation for the effects/limitations of this retrograde flow is a cornerstone of both the selection and management of VA ECMO. We will review the details of retrograde flow in detail in the next chapter.

PUTTING IT TOGETHER: VA ECMO AND THE VICIOUS CYCLE OF CARDIOGENIC SHOCK

In this chapter, we took a closer look at the vicious cycle of cardiogenic shock, getting a better sense of how the effects can spiral out of control. Rather than simply the consequence of low cardiac output, the cycle is the result of the compounding effects on multiple phenomena. The confluence of these effects is what dictates the rationale for VA ECMO, particularly, for which patients would be well suited. The higher the anticipated benefits and the lower the limitations will dictate the degree that VA ECMO will be of benefit to any individual patient. Let's review the effects that we discussed with attention to the effect ECMO will have:

✓	Decreased cardiac output
✓	Decreased coronary perfusion
⃠	Increased LV distension
✓	Hypoxia → coronary hypoxia
✓	Acidosis → vasodilation
✓	Acute kidney injury/fluid retention
✓	Worsening RV afterload
✓	Increased RV distension
✓	Pressors/fluid worsening RV

You will notice that VA ECMO has a positive effect on the majority of phenomena associated with cardiogenic shock, with the notable exception of increased left ventricular distension, which can potentially be worsened due to retrograde flow. Understanding retrograde flow is essential to understand the effect of deploying VA ECMO in this patient population, which is why we will dedicate the next chapter solely to the physiology of this phenomenon.

SUGGESTED READING

Beard, D. A., & Feigl, E. O. (2011). Understanding Guyton's venous return curves. *American Journal of Physiology-Heart and Circulatory Physiology*., *301*(3), H629–H633.

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